

Mentor Engineers — Comprehensive Issues Analysis & Fix Action Plan

Date: 2026-09-22

Target Environment: Local Development & Live Production VPS

Affected Components:

- Supabase Edge Function: `supaba Hari`   `se/functions/ai-chat-mentor-engineeros/index.ts`
- Frontend Dashboard: `frontend/src/components/DetailingCalendar.tsx` & `frontend/src/pages/Orders.tsx`
- Docker Deployment: `docker compose` & Edge Runtime container

Executive Summary

Through rigorous investigation and local simulation using the live AI Gateway and date parsing utilities, we identified **5 core technical root causes** for the issues observed on the live server:

Visible Problem	Technical Root Cause	Impact
1. Calendar shows booking on wrong date (e.g. today)	Date format mismatch (<code>September 24th</code> / <code>24/09/2026</code> instead of <code>YYYY-MM-DD</code>) causing Moment.js in <code>DetailingCalendar.tsx</code> to fail parsing and fallback to <code>o.created_at</code> (today).	Bookings do not appear on the requested appointment date in the calendar view.
2. Booking details not showing in Calendar	(a) Invalid date strings fall back to today. (b) Strict category filter in <code>DetailingCalendar.tsx</code> discards orders without exact match. (c) CSS cell height clipping in <code>react-big-calendar</code> . (d) Order wasn't confirmed explicitly, so <code><ORDER_JSON></code> was never written.	User navigates to the booked date on the calendar and sees nothing or clipped details.
3. Double booking on already booked dates	<code>bookedDates.includes(dateStr)</code> looks for exact <code>YYYY-MM-DD</code> strings. Non-standard dates (<code>"September 24th"</code>) evaluate to <code>false</code> , allowing AI re-booking.	Multiple customers can book detailing on the exact same date.
4. AI not suggesting available free dates on Live Server	(a) Sequence Lock prompts packages first. (b) Customer specifies date first. (c) Add-on request bypassed trigger. (d) VPS Docker cache ran old code.	Customers aren't guided to select one of the 3 pre-calculated dates.
5. Timezone discrepancy	Supabase Edge Function runs in UTC. Sri Lanka is UTC+5:30. <code>new Date()</code> generates yesterday's date for early morning hours.	"Tomorrow" and next 3 days are calculated with an offset day error.

Detailed Analysis of Issues

Issue A: Live Server-ல் Booking Details Calendar-ல் காட்டாததற்கான முழுமையான காரணங்கள்

We verified this behavior locally by simulating orders with different data structures. The booking details failed to show for 4 distinct reasons:

1. Date Fallback to "Today" (The booked date looks completely empty)

When an order is created, if the AI saves the date as "September 26th" or "26/09/2026" :

```
let start = moment(`${dateStr}T08:30:00`).toDate(); if (isNaN(start.getTime())) { start = moment(o.created_at).set({ hour: 8, minute: 30, second: 0 }).toDate(); end = moment(o.created_at).set({ hour: 17, minute: 30, second: 0 }).toDate(); }
```

`moment("September 26thT08:30:00")` is **Invalid Date**.

`isNaN(start.getTime())` evaluates to **true**.

The system automatically assigns `start` to `o.created_at` (today).

Result: When navigating to September 26 on the calendar, the date is 100% EMPTY! The booking is mistakenly placed on today's date.

2. Strict Service Category Filter in `DetailingCalendar.tsx`

In `DetailingCalendar.tsx` :

```
const parsedEvents = orders .filter(o => { const cf = o.custom_fields; if (!cf) return false; const cat = (cf.service_category || '').toLowerCase(); return ( cat === 'detailing' || cat === 'detailing & polish' || (Array.isArray(cf.add_ons) && cf.add_ons.some((a: string) => a.toLowerCase().includes('detail')))) ); })
```

If the AI saved `service_package = "Interior Detailing"` , but left `service_category = ""` or set it to `"mechanical"` , this filter returns **false**. The order is completely dropped.

3. CSS Clipping & Month View Overflow in `react-big-calendar`

The `CustomEvent` component requires at least `110px` of vertical height to display Name, Phone, Plate, Package, and Price. In default month view, rows have fixed height (~80px) and `overflow: hidden;` . The event card is either clipped or collapsed into a small `+1 more` link.

4. The Order Was Never Written to DB

In `ai-chat-mentor-engineeros/index.ts` :

```
DO NOT output <ORDER_JSON> when presenting the booking summary or asking for confirmation! Output <ORDER_JSON> ONLY AFTER the customer explicitly replies confirming the booking (e.g. "Yes", "Confirm", "ஐ", "ஓ").
```

Issue B: Double Bookings on Already Booked Dates

In `ai-chat-mentor-engineeros/index.ts`:

```
const bookedDates = (openOrders || []) .filter(...) .map(o => { const cf =  
o.custom_fields as Record<string, any>; return cf?.booking_date || (o.created_at ?  
o.created_at.split('T')[0] : null); }) .filter(Boolean); const isDateAvailable = (date: Date)  
=> { const dateStr = formatDate(date); // Output: "YYYY-MM-DD" if  
(bookedDates.includes(dateStr)) return false; // Booked check ... };
```

If an order was previously saved with `booking_date = "September 24th"`, `bookedDates` contains `["September 24th"]`. When testing if `"2026-09-24"` is available, `["September 24th"].includes("2026-09-24")` evaluates to **false**. The system believes the date is vacant and freely offers it again.

Issue C: Live Server Not Suggesting Available Free Dates

1. **Multi-Turn Sequence Lock:** The prompt enforces asking for package details and add-ons first. Dates are not presented until the customer explicitly replies to the add-on inquiry.
2. **Customer Supplies Date Upfront:** If the customer opens with *"Can I book detailing for tomorrow?"*, the AI previously lacked a strict rejection rule, acknowledging the user's date without offering the pre-calculated ones.
3. **Detailing as an Add-On:** If a customer requested "Full Service" and picked "Cut & Polish" as an add-on, the AI omitted detailing date restrictions.
4. **VPS Docker Container Caching:** The `edge-runtime` container caches compiled TypeScript in Deno memory. When updating code via `git pull`, the container runs the old cached version until explicitly restarted.

Issue D: Timezone Mismatch (UTC vs Sri Lanka UTC+5:30)

The live VPS operating system runs in standard UTC time.

```
const todayStr = new Date().toISOString().split('T')[0]; let checkDate = new Date();  
checkDate.setDate(checkDate.getDate() + 1);
```

Between 12:00 AM and 05:30 AM in Sri Lanka, the server's UTC time is still yesterday. This causes `todayStr` to be 1 day in the past, and `checkDate` (+1 day) evaluates to *today* instead of *tomorrow*.

Step-by-Step Fix Action Plan

1. Fix Calendar Event Rendering & Parsing (`DetailingCalendar.tsx`)

- **Multi-Format Date Parser:**

```
const dateStr = o.custom_fields?.booking_date; let parsed = moment(dateStr, ['YYYY-MM-DD', 'YYYY/MM/DD', 'DD-MM-YYYY', 'DD/MM/YYYY', moment.ISO_8601], true); if (!parsed.isValid()) parsed = moment(dateStr); const finalMoment = parsed.isValid() ? parsed : moment(o.created_at); let start = finalMoment.clone().set({ hour: 8, minute: 30, second: 0 }).toDate();
```

- **Robust Filter:** Match not only `service_category === 'detailing'`, but also check `service_package` (e.g. contains 'detail' or 'polish') or order items.
- **Popup & Height Enhancements:** Add `popup={true}` and increase row styling in `react-big-calendar` so events don't clip and all details are cleanly visible.

2. Enforce Strict `YYYY-MM-DD` Date Storage in Backend

- **Prompt Schema Definition:** Update `<ORDER_JSON>` to strictly require `YYYY-MM-DD` (e.g. `2026-09-24`).
- **Backend Normalization Function:** Before saving into `orders.custom_fields`, run the date through `normalizeToISODate()` to guarantee formatting.
- **Booked Dates Retrieval:** Normalize all existing orders in the DB during query so `bookedDates.includes("YYYY-MM-DD")` never fails.

3. Timezone Fix for Sri Lanka

Calculate `todayStr` and `checkDate` using UTC+5:30 offset so that late-night and early-morning bookings are always in sync with Sri Lankan business days.

4. Date Suggestion Prompt Lock

Explicitly instruct the AI that ANY detailing request (main or add-on) MUST suggest the 3 pre-calculated dates. If a customer suggests any other date, explicitly reject it. Provide clean English and Sinhala templates so the AI never drops dates during language switching.

5. VPS Deployment Procedure

```
git pull origin master docker compose restart functions docker compose ps functions
```